Event-Driven Analysis of Soccer Game Dynamics

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1. Introduction

1.1 Project Scope

In the realm of soccer, a sport known for its dynamic nature, key events such as goals, card issuances, and penalty awards can significantly alter the trajectory of a match. These pivotal moments often trigger shifts in various game metrics, including ball possession rates, successful pass counts, and attacking play frequencies. Employing quality engineering principles, particularly the use of control charts, presents a robust methodology for monitoring these soccer statistics and identifying "out-of-control" situations, thereby offering valuable insights into the intricacies of game dynamics.

1.2 Research Aims

The core objectives of this investigation are as follows: To implement statistical quality control techniques in the analysis of soccer match data, conducting analyses of fluctuations in game statistics, assess the efficiency of various detection strategies for identifying these fluctuations, establish correlations between observed statistical shifts and specific in-game events and investigate the feasibility of developing early warning indicators for impending significant match events.

1.3 Data Overview

The study utilizes a comprehensive dataset comprising minute-by-minute statistics from 102 distinct soccer matches. Each data entry corresponds to a specific minute of gameplay and encompasses the following key parameters:

- fixture_id: A unique identifier assigned to each individual match
- minute: The specific minute of gameplay being recorded
- Successful.Passes_home / away: Quantification of successful passes executed by home and away teams
- Ball.Possession_home / away: Percentage breakdown of ball possession between home and away teams
- Goals_home / away: Goal tallies for both home and away teams
- Yellowcards_home / away, Redcards_home / away: Record of disciplinary cards issued to both teams
- Penalties_home / away: Documentation of penalties awarded to each team
- Attacks_home / away: Enumeration of attacking plays initiated by both home and away teams

This rich dataset provides a multifaceted view of match progression, enabling in-depth analysis of how various events influence game dynamics over time.

Task 1: Descriptive Analysis of Shifts in Statistics

2.1 Data Preparation

Given that some of the statistics in the dataset are results of cumulative actions over time (e.g., possession, passes), I proposed and implemented three different transformation approaches to adequately monitor these variables and examples of their outputs.

2.1.1 Attacks per Minute Ratio

This approach considers the attacks per minute ratio for each match instance. The motivation behind using this ratio is the assumption that each team has an average number of attack plays per minute, and this rate remains close to the average throughout most of the match unless there is a disrupting event.

- Attacks_home_per_min = Attacks_home / minute

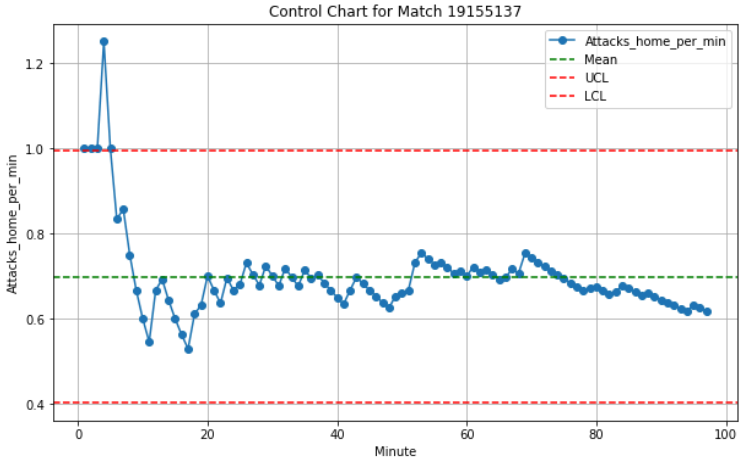


Figure 1

- Attacks_away_per_min = Attacks_away / minute

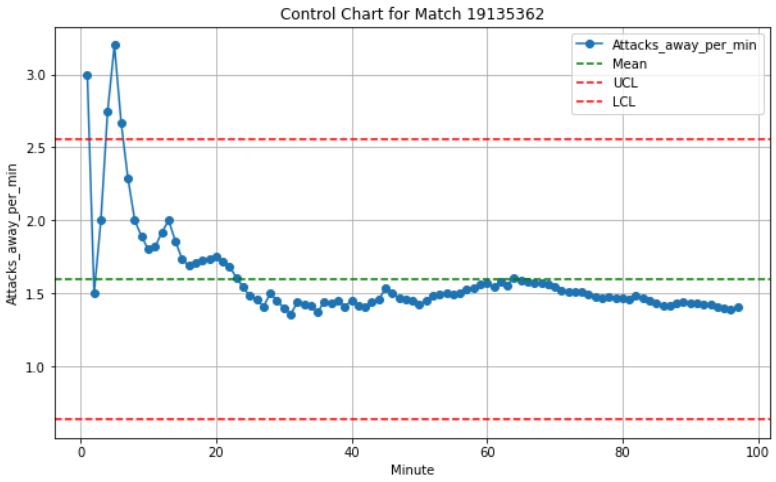


Figure 2

2.1.2 Incremental Changes in Attacks (Differencing)

This approach focuses on the incremental change in attack plays of the teams. The assumption is that the number of attacks in each minute of the game should be comparable and close.

Attacks_home_differenced = Attacks_home.diff().fillna(0)

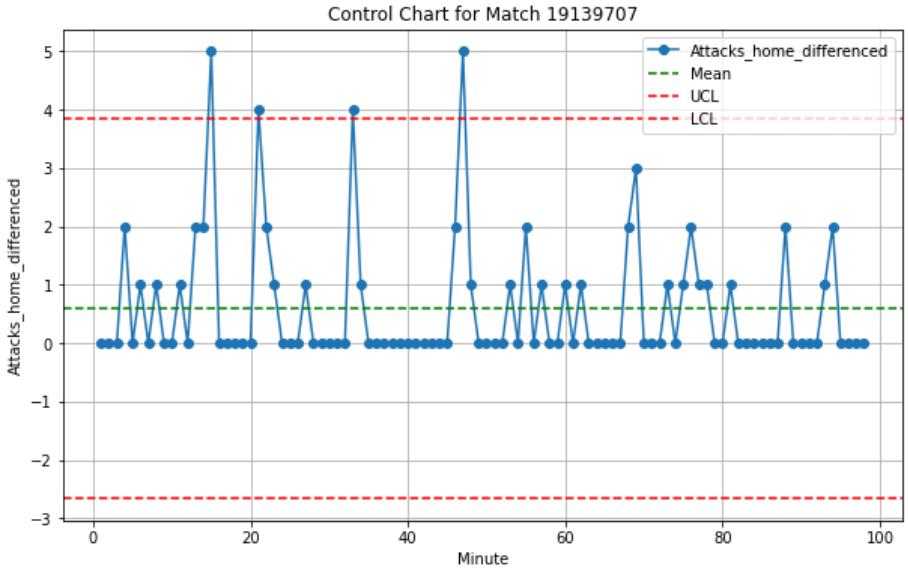


Figure 3

Attacks_away_differenced = Attacks_away.diff().fillna(0)

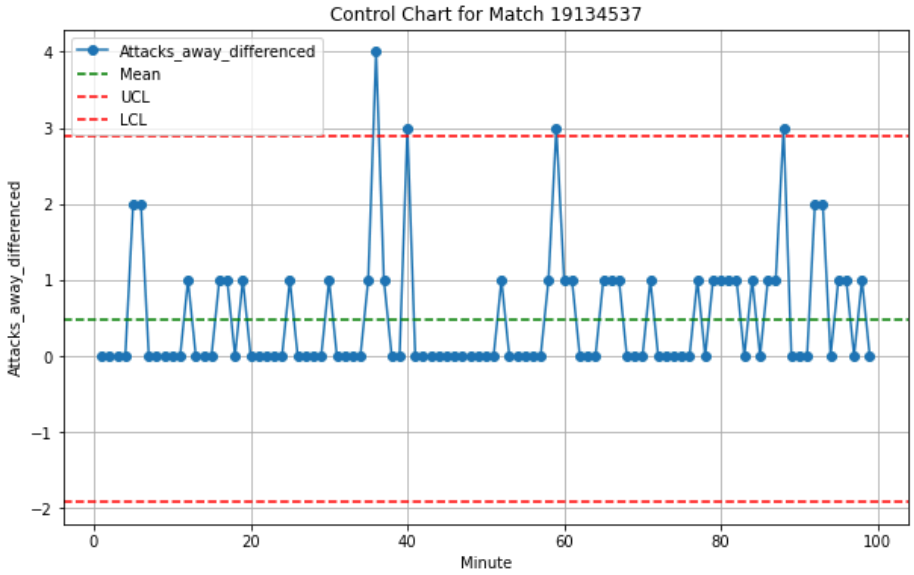


Figure 4

2.1.3 Normalized Attack Difference

This approach involves the attack plays of both teams to understand whether there is an unusual event affecting the dominance of the home team over the away team in terms of attacking rates. The objective is to monitor relative performance instead of focusing on a single team, aiming to detect disruptions in the game more effectively.

Attacks_diff_normalized = (Attacks_home - Attacks_away) / (Attacks_home + Attacks_away)

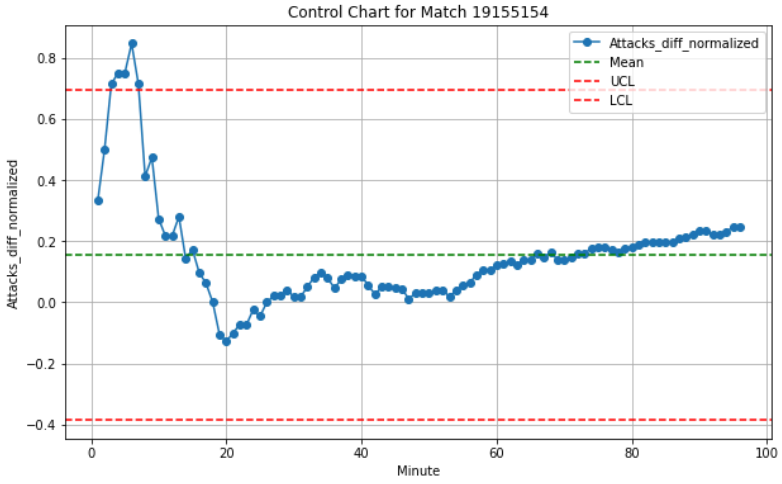


Figure 5

2.2 Control Chart Construction

For each of the transformed metrics, Shewhart control charts using 3-sigma limits are constructed. The control limits were calculated as follows:

- Upper Control Limit (UCL) = mean + 3 * standard deviation
- Lower Control Limit (LCL) = mean - 3 * standard deviation

These control charts were used to monitor the selected metrics for "out-of-control" conditions throughout each match.

2.3 Out-of-Control Detection Methods

Several functions to detect and analyze out-of-control situations have been developed:

- detect_out_of_control: This function identifies instances where the metric exceeds the control limits and records the minute and direction (above UCL or below LCL) of the exceedance.
- plot_control_charts: This function visualizes the control charts for specific matches, highlighting the mean, UCL, and LCL.
- identify_out_of_control_and_plot: This function combines the detection of out-of-control situations with the plotting of control charts for the top 8 matches in terms of total number of out-of-control situations.

2.4 Event Analysis Approach

To establish connections between detected shifts and preceding game events, a structured approach was implemented. This methodology involved examining events that occurred within a 7-minute window prior to each out-of-control situation. These events were then categorized based on their type (such as goals, yellow cards, red cards, and penalties) and the team involved (home or away). A specialized function, named check_total_events_before_ooc_wrt_sign, was developed (See Appendix A.2) to aggregate the total number of each event type occurring before out-of-control situations. The results were further differentiated based on whether the situation was above the Upper Control Limit (UCL) or below the Lower Control Limit (LCL). Additionally, a focused analysis was conducted specifically for goals scored when the match was tied. This involved filtering out-of-control instances where a goal was scored in the preceding 7 minutes of a tied game, followed by plotting control charts for these specific instances to visualize the goal's impact on attack patterns. This comprehensive methodology enables a thorough analysis of how various events in a soccer match can trigger shifts in attack metric and allows for the assessment of how significantly these shifts can be detected using statistical quality control methods.

3. Results and Analysis

3.1 Comparison of Transformation Approaches

After implementing the three transformation approaches (Attacks per Minute Ratio, Incremental Changes in Attacks, and Normalized Attack Difference), I observed the following:

- Attacks per Minute Ratio and Normalized Attack Difference: These approaches showed similar patterns in their control charts. All out-of-control situations were detected in the first minutes of the games. This indicates that these out-of-control situations don't reflect meaningful results or represent actual out-of-control situations, but rather reflect noisy data at the beginning of the games. This is due to the way these metrics are calculated, which inherently leads to large deviations from the mean in the early stages of a match.
- Incremental Changes in Attacks (Differencing): This approach showed more promising results, with out-of-control situations not particularly concentrated at the beginning of the games. However, all of the out-of-control situations occurred above the UCL because attack plays cannot decrease, they can only remain the same or increase. Based on these observations, a decision was made to adopt a modified approach for the differencing method. The difference between the series of differenced attacks of the home team and away team was calculated, resulting in the creation of a new metric named "Differenced_attacks_diff". This metric demonstrated a more balanced distribution of out-of-control situations, with occurrences observed both below the LCL and above the UCL.

3.2 Analysis of Out-of-Control Situations

3.2.1 Overall Statistics

Using the "Differenced_attacks_diff" metric, Out-of-control situations across all matches have been analyzed. The results were as follows:

For +UCL (Upper Control Limit) situations:

- Total occurrences: 35
- Events in the preceding 7 minutes:
    - 5 home goals
    - 5 away goals
    - 1 home yellow card
    - 5 away yellow cards
    - 0 red cards
    - 1 away penalty

For -LCL (Lower Control Limit) situations:

- Total occurrences: 42
- Events in the preceding 7 minutes:

- 3 home goals
- 3 away goals
- 6 home yellow cards
- 5 away yellow cards
- 0 red cards
- 0 penalties

3.2.2 Event Distribution Before Out-of-Control Instances

The distribution of events before out-of-control situations provides interesting insights:

- Goals: Both home and away goals appear to have a similar impact on causing out-of-control situations, with a slight tendency towards +UCL situations.
- Yellow Cards: Yellow cards seem to have a more pronounced effect on -LCL situations, particularly for the home team. This could suggest that receiving a yellow card might lead to more cautious play, reducing the attack differential.
- Red Cards: Interestingly, no red cards were observed in the 7-minute window before out-of-control situations. This could be due to the rarity of red cards or suggest that their impact might take longer to manifest in attack patterns.
- Penalties: Only one away penalty was associated with a +UCL situation, indicating that penalties might not have an immediate significant impact on attack patterns or there are too few penalty occurrences in the dataset for the analyzed methodology to detect such a small sample.

3.3 Goal Impact Analysis

The hypothesis/relation is if a team scores a goal when the match is tie/draw, this goal tends to increase the scoring team's relative attack plays because the other team would become more cautious and play defensively as their game system would get more fragile and be disrupted after the goal. In other words, they would first focus on recovering, mitigating the effect of the setback and avoiding risky plays since they fell behind in the score. An additional goal/mistake could potentially push them out of the match and make the comeback more difficult. On the other hand, it is expected for the scoring team to be more comfortable in increasing their attack plays with the momentum gained from the goal. They would be looking to take advantage of the opponent's disorganization to control the game and create more goal opportunities with confidence before the opponent team recovers. Thus, they would play more offensively and their attacks would increase compared to the opponent's attacks in the first minutes following the goal.

3.3.1 Home Team Goals

Three instances were identified where a home goal was scored in a tied game before an out-of-control situation. Mixed results were shown by the control charts for these matches:

- Match ID 19135260: A goal at minute 40 led to a +UCL situation at minute 45, supporting the hypothesis.

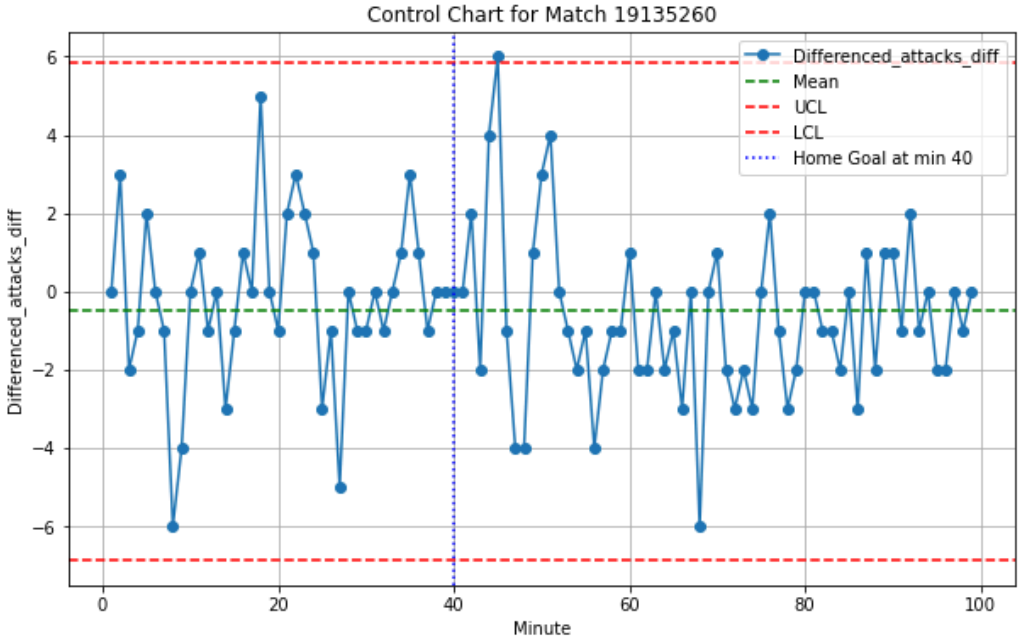


Figure 6

- Match ID 19135283: A goal at minute 24 led to a +UCL situation at minute 30, also supporting the hypothesis.

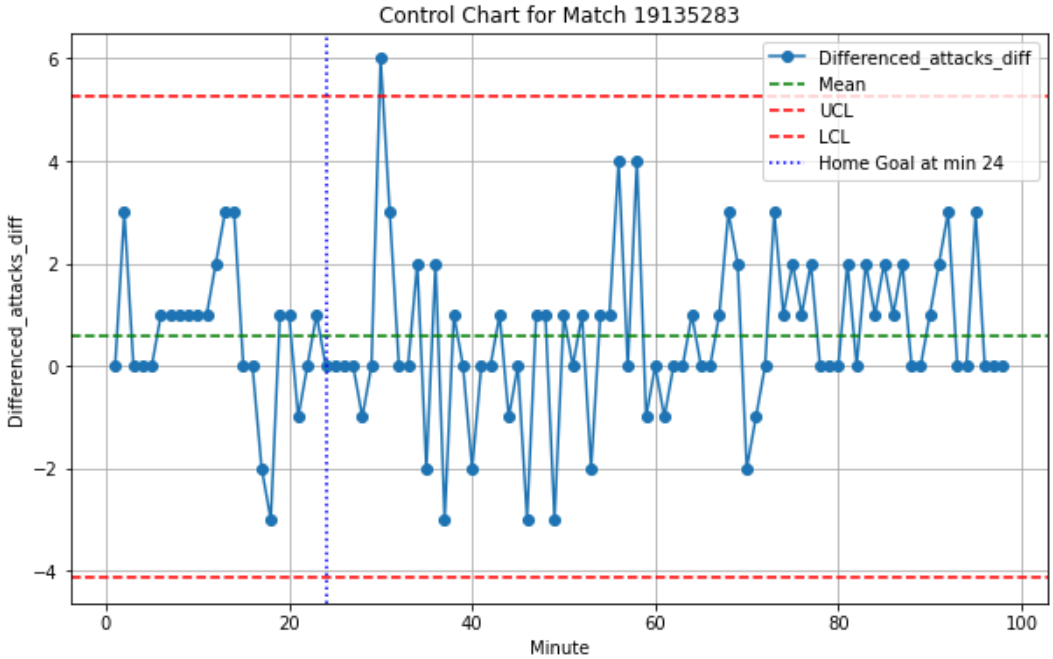


Figure 7

- Match ID 19139711: A goal at minute 73 led to a +UCL situation at minute 78, also supporting the hypothesis

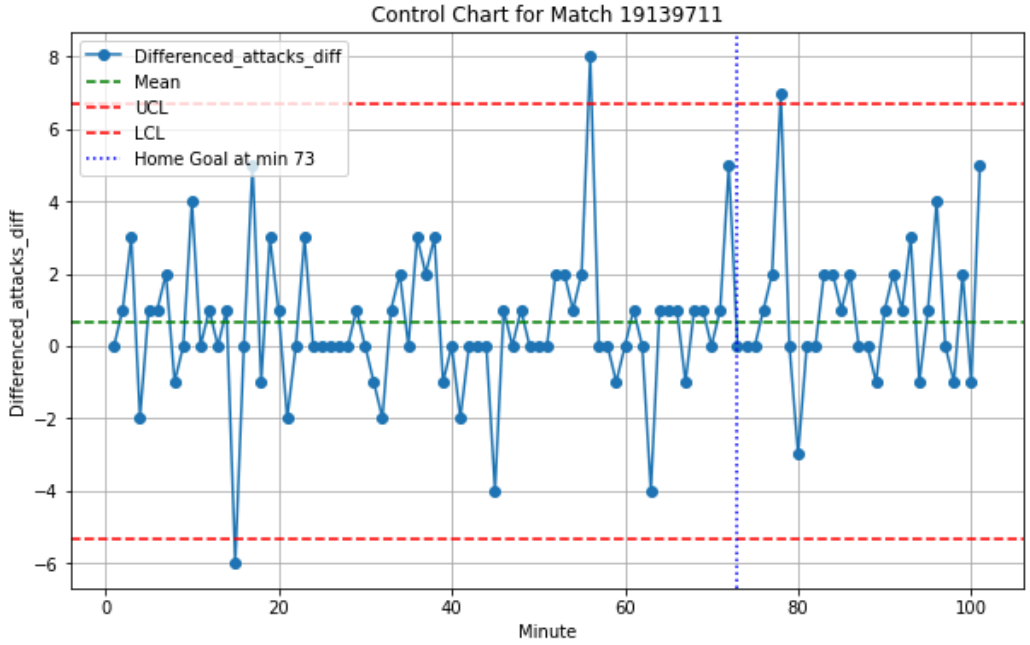


Figure 8

- Match ID 19155090: A goal at minute 4 led to a -LCL situation at minute 6, contradicting the hypothesis. However, this early-game situation might not be representative due to the volatility of play in the opening minutes. Hence this case will be disregarded.

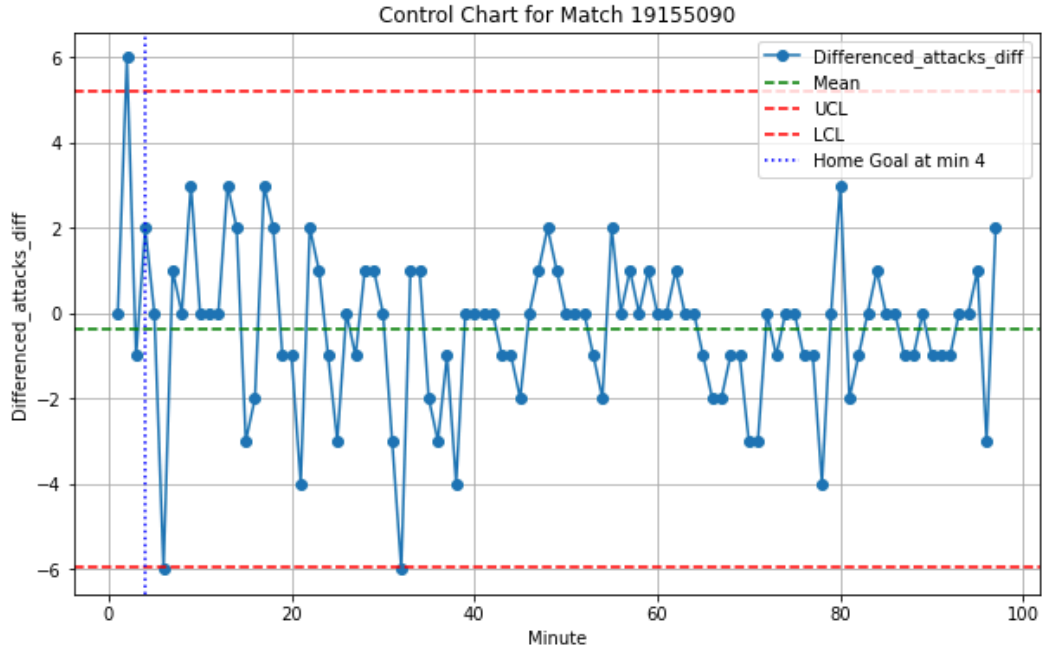


Figure 9

- Match ID 19155151: A goal at minute 21 led to a + UCL situation at minute 28, supporting the hypothesis.

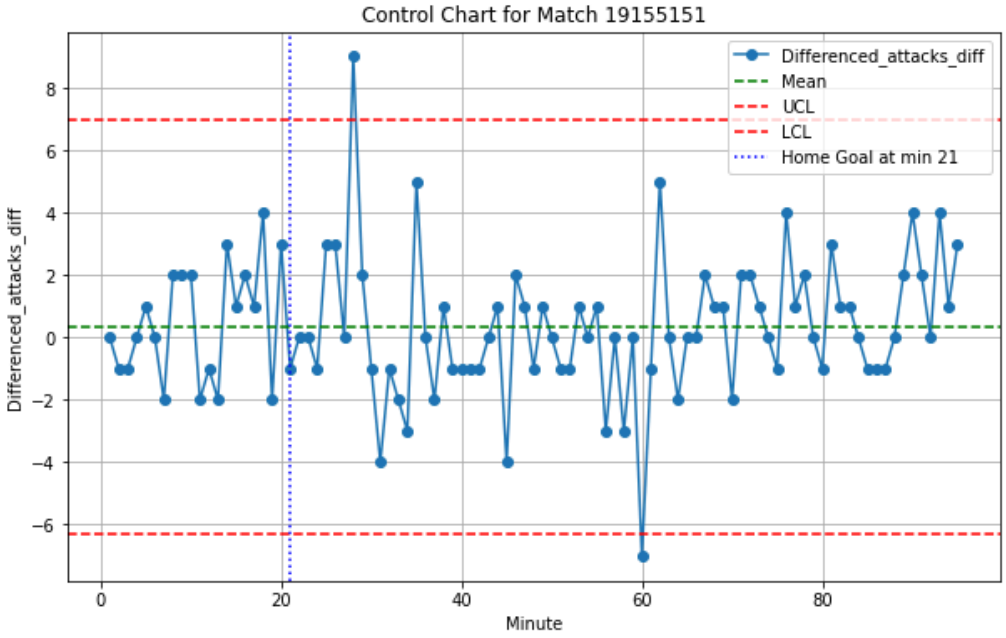


Figure 10

3.3.2 Away Team Goals

Two instances of away goals scored in tied games before out-of-control situations were identified:

- Match ID 19155160: A goal at minute 53 led to a -LCL situation at minute 55, supporting the hypothesis that an away goal might decrease the home team's relative attack rate.

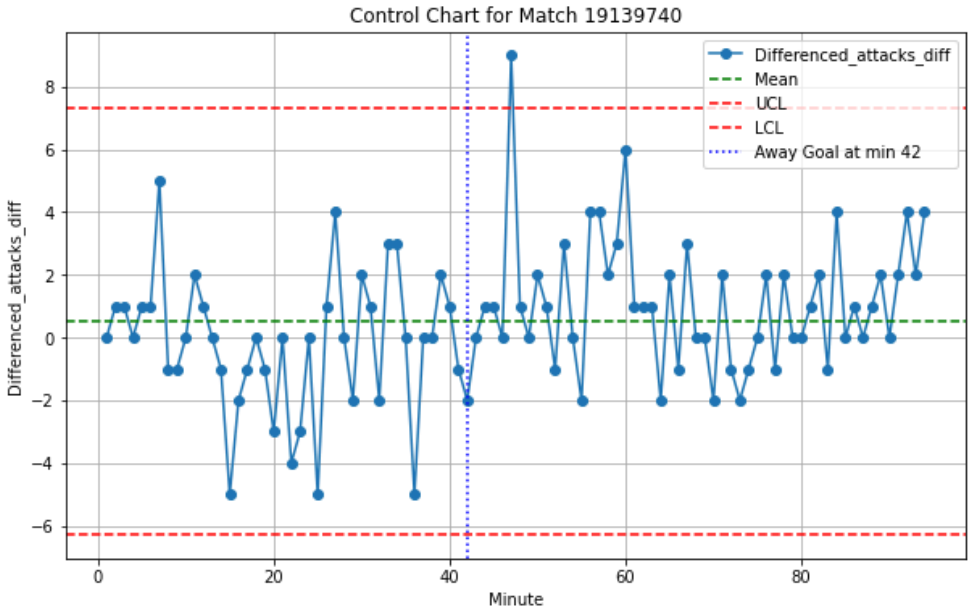


Figure 11

- Match ID 19139740: A goal in the first half led to a +UCL situation in the second half. This case was discarded from the analysis as the time gap between the goal and the shift was too large to infer a direct relationship.

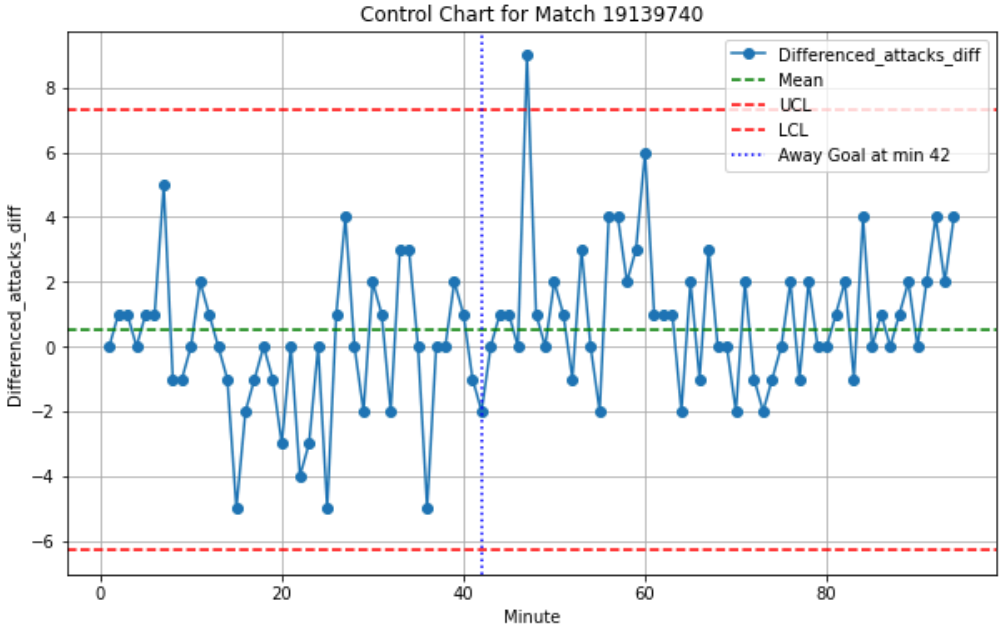


Figure 12

3.4 Case Studies

3.4.1 Detailed Analysis of Selected Matches

After analyzing part 3.3 and eliminating the two cases that could mislead the results—due to occurring too early in the game and the points being in different halves of the soccer match—five matches showed a similar pattern. Therefore, it can be concluded that when the game is tied, a goal scored by one team causes a shift in the attack metric, as observed in these five matches. From this set of five matches, two were selected for a case study to conduct a detailed analysis of the matches that exhibited clear shifts in attack patterns.

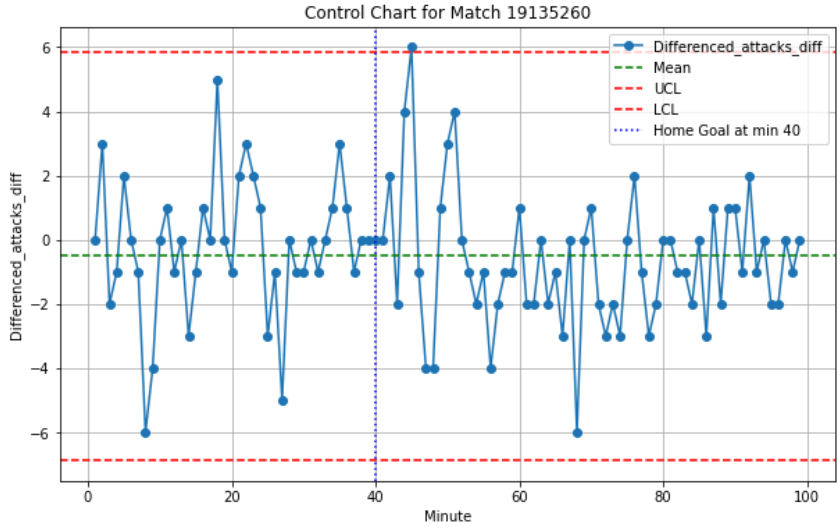


Figure 13

- Home goal scored at minute 40 when the score was 0-0
- +UCL situation detected at minute 45
- The control chart shows a clear increase in the attack differential favoring the home team immediately after the goal

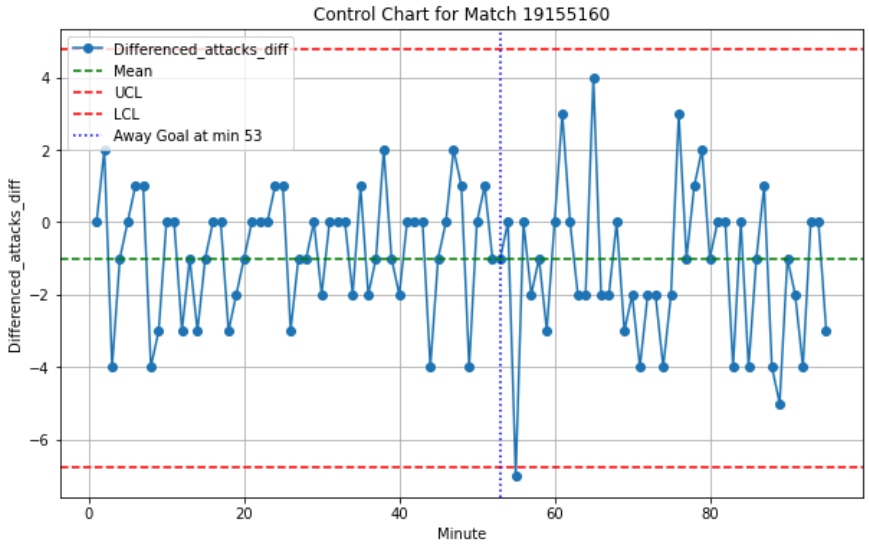


Figure 14

- Away goal scored at minute 53 when the score was 0-0
- -LCL situation detected at minute 55
- The control chart shows a sharp decrease in the attack differential, indicating a reduction in home team attacks relative to the away team

Finally, to evaluate how quickly the control charts detect mean shifts, a metric called 'time until detection' is used for evaluation. For this metric, the selected five matches are analyzed, and the time difference between the goal event and the OOC point is calculated. The average time is then computed. From the calculation, the average time until detection is found to be 5 minutes for the 5 matches that have a similar pattern.

4. Discussion

4.1 Interpretation of Results

The analysis of out-of-control situations in soccer matches using the "Differenced_attacks_diff" metric has revealed several interesting patterns:

- Impact of Goals: The results suggest that goals do have a significant impact on attack patterns, especially when scored in tied games. However, this impact is not uniform across all matches. In some cases, a goal leads to an increase in attacks by the scoring team (as seen in the Manchester United vs Fulham match), while in others, it may lead to a decrease in attacks by the conceding team (as observed in the match with ID 19155160).
- Yellow Cards: The higher number of yellow cards associated with -LCL situations, particularly for home teams, suggests that receiving a yellow card might lead to more

cautious play. This could be interpreted as teams becoming more defensive after a player is cautioned, potentially to avoid further disciplinary action.

- Lack of Red Card Impact: The absence of red cards in the 7-minute window before out-of-control situations is notable. This could be due to the relative rarity of red cards or suggest that teams might take longer to adjust their tactics following a sending-off.
- Penalties: The limited impact of penalties on attack patterns immediately after they occur is somewhat surprising. This might indicate that teams do not significantly alter their attacking approach immediately after a penalty.

4.2 Implications for Game Dynamics

The findings of this study provide valuable insights into the dynamics of soccer matches. The analysis highlights the significance of momentum shifts, particularly following goals scored in tied games, suggesting that such moments can be pivotal in determining match outcomes. These insights could aid coaches and analysts in developing strategies to leverage or counteract these shifts effectively. Additionally, the impact of disciplinary actions was observed to vary; while yellow cards tend to have an immediate influence on play, the consequences of red cards appear to unfold more gradually as teams adapt their tactics to compensate for the loss of a player. Furthermore, the observed differences in patterns based on match context, such as tied versus non-tied states, underscore the importance of situational factors in soccer analytics. These findings collectively contribute to a deeper understanding of the multifaceted factors shaping soccer match dynamics.

4.3 Effectiveness of Detection Methods

The "Differenced_attacks_diff" metric proved to be the most effective in detecting meaningful shifts in attack patterns. This metric successfully addressed the limitations of the other approaches:

- It avoided the noise issues at the beginning of matches that affected the Attacks per Minute Ratio and Normalized Attack Difference metrics.
- It allowed for both positive and negative out-of-control situations, unlike the simple differencing approach, which only showed positive deviations.
- It also accounted for the correlation between the "attacks home" and "attacks away" metrics. This correlation is crucial as it reflects the relative performance of the teams, making it easier to detect out-of-control situations. Analyzing these metrics separately could hide this relative relationship and lead to misleading results.

The use of Shewhart control charts with 3-sigma limits provided a clear visual representation of shifts in attack patterns. The selection of a 7-minute window for examining events before out-of-control situations was based on a balance between having a sufficient sample size and avoiding overly large intervals, such as 15 minutes, which may dilute the data's sensitivity to

changes. Similarly, smaller intervals, such as 5 minutes, might not provide enough data points for reliable analysis. While the 7-minute window was determined to be the most feasible option for this study, future research could explore alternative interval lengths to further refine the analysis.

4.4 Limitations of the Study

The study faced several limitations that should be considered when interpreting the results. While the dataset encompassed 102 matches, the occurrence of specific scenarios, such as goals in tied games, was relatively infrequent, potentially limiting the generalizability of certain findings. The complex nature of soccer means that numerous factors beyond those captured in the dataset could influence attack patterns, including team strength, league context, and specific tactical approaches, which were not accounted for in this analysis. The selection of a 7-minute window for examining events preceding out-of-control situations, while practical, may not have captured all relevant precursors to shifts in attack patterns. Additionally, although the "Differenced_attacks_diff" metric proved valuable, it may not fully encapsulate all nuances of attack patterns, such as the quality of attacks or possession-based strategies. Lastly, the current study's focus on descriptive analysis of past events does not address the potential for using these methods to predict future events or shifts in game dynamics, leaving room for further exploration in predictive analytics within this context.

5. Key Findings

The study yielded several key findings that provide insights into soccer match dynamics. The "Differenced_attacks_diff" metric, which represents the difference between home and away team's incremental attacks, emerged as the most effective tool for detecting meaningful shifts in attack patterns during matches. It was observed that goals scored in tied games often led to significant changes in attack patterns for 5 matches in total, although the nature of these shifts varied, sometimes increasing and other times decreasing the relative attack rate. Yellow cards were found to have an immediate impact on play, frequently resulting in more cautious approaches, particularly for home teams. Interestingly, the impact of red cards and penalties on immediate attack patterns was less pronounced than initially hypothesized, suggesting a need for further investigation into their long-term effects. Overall, the study demonstrated the significant potential of using statistical quality control methods, especially control charts, in analyzing and understanding the complex dynamics of soccer matches since by using the methodology presented 5 mean shifts are successfully detected in an average of 5 minutes.

Task 2: Preventive Analysis Using Warning Sign

1. Introduction

In this task, the aim is to analyze a selected metric to foresee significant future events. For this purpose, the requirement is to focus on patterns in the match data to find a pattern that can be related to a significant event as its cause. The attack metric, both for the home and away teams, is chosen, and analyses are made on this metric's ability to give an early warning signal for an event in the match.

For the aim of finding the relationship between out-of-control points and the events coming afterward, analyses started with data manipulation, deciding the intervals, finding the out-of-control points, and then analyzing the patterns in these points to see if there is a reliable source of information during a match that can be a significant estimator for events.

These analyses are followed by a reliability analysis that calculates how significant the findings are, along with a brief conclusion that summarizes the results.

2. Methodology

The methodology for Task 2 had three main points: selection of metrics and data manipulation; decision of interval settings and finding out-of-control points; and finally, finding the pattern and creating a hypothesis that its reliability would be calculated afterward.

First, due to my hypothesis that it could have the most significant relationship with goal events, the attack metric was chosen. However, the match data being examined was cumulative, so it required some manipulation. To eliminate the cumulative behavior, the difference was taken. Additionally, the data included the attack metrics for both the home and away teams. Instead of analyzing them separately, I decided to use their difference as the metric. These two metrics are highly correlated, and not accounting for the relationship between them could result in false interpretation. Therefore, the metric used in the control chart was defined as "Attacks_home - Attacks_away," which will be referred to as "Attacks_difference" in the further analysis.

After the required data manipulation, the second step was to determine how to approach the examination of match data and on which intervals. This was necessary because the aim of this task was to foresee upcoming events, and using the entire match's data was not feasible. The goal was to, after collecting some data, have an idea about the future events. Hence, the match was examined in intervals. The approach was to look at match data in 15-minute rolling intervals. This meant taking the first 15 minutes and then adding the next 16th minute while deleting the first minute considered, and checking if the addition of this last minute resulted in an out-of-control point. This approach was chosen for two reasons: first, to account for each minute

in the match, and second, to discard data that could distort the forecast over time. In a soccer game, what happens in the first 5 minutes loses importance as the match progresses, so considering that in each interval would add too much noise to the analysis. After this decision, out-of-control points occurring after each new data addition to the interval were found, and events happening 8 minutes after these out-of-control points were stored in a table for further analysis. The 8-minute forecasting window was chosen because, within 10 minutes, many events can occur, making it complicated for the analysis to assign a cause. An interval such as a 5-minute window was not used because there would be too few data points, resulting in an insufficient sample size for analysis.

Lastly, by examining the table containing the out-of-control points with the events happening in the chosen window, hypotheses were drawn. To support these hypotheses, matches with similar patterns were identified. For the matches where the situation contradicted the hypothesis, match data was examined one by one to see if there could be another assignable cause affecting the result. Finally, for my hypothesis, metrics such as time until detection and false alarm rate were calculated to evaluate the results.

3. Analysis Method

The matches data contains cumulative data difference for the selected metrics Attacks_home and Attacks_away are taken. Since the Attacks_home and Attacks_away metrics are correlated, analysing them separately would result in misleading conclusions. Hence their difference is taken and analysis is continued by using the Attacks_difference metric.

Attacks_difference = Home_Attack_Change - Away_Attack_Change

The events data is also stored in the matches dataset cumulatively. Since the aim is to examine whether a change occurred or not for the given metric, events data is converted to 1 and 0's. 1 meaning that in the given minute a change occurred in that event and 0 indicating that no change occurred in that minute.

4. Out of Control Points Analysis

The match data is examined in 15-minute rolling intervals with the aim of having enough sample points and eliminating past data that may cause misleading information about the statistics of the match data. This approach results in dynamic control charts where the 3-sigma control limits are used.

Out-of-control points that occur due to the addition of new data in these rolling intervals are stored along with the events that happen 8 minutes after the out-of-control point. The time difference between the OOC point and the events is also stored as a metric. Additionally, out-of-control points are classified as upper and lower in order to identify the direction of the

mean shift and categorize the out-of-control points with similar patterns in terms of shift direction.

This exercise is performed for each match using the function called "Analyze_Attack", which returns a list of outliers, including the out-of-control points. The related code is given in Appendix A.1.

Each unique match is analyzed using the Analyze_attack function. For each match, the minutes where OOC points are found are marked as outliers. For ease of analysis, these outlier points and the related data attached to them are provided in a data frame as can be seen from Figure 15.

	fixture_id	Type	Minute	Goals_home_Change	Goals_home_Change_T	Goals_away_Change	Goals_away_Change_T	Yellowcards_home_Change	Yellowcards_home_Change_T	Yellowcards_away_Ch
0	19172022	Upper	67	0	0	0	0	0	0	
1	19172022	Upper	87	1	2	0	0	0	0	
2	19135283	Upper	30	0	0	0	0	0	0	
3	19135298	Upper	86	0	0	0	0	1	0	
4	19134492	Lower	79	0	0	0	0	1	3	
5	19154573	Lower	34	0	0	0	0	0	0	
6	19155137	Upper	52	0	0	1	5	0	0	
7	19155137	Lower	74	1	2	0	0	0	0	
8	19139719	Lower	51	0	0	0	0	0	0	
9	19139735	Lower	45	0	0	0	0	0	0	
10	19139736	Lower	33	0	0	0	0	0	0	
11	19154632	Upper	52	0	0	0	0	1	7	

Figure 15

From the above table, I have 12 OOC points occurring in 10 unique matches. After the OOC points, the control charts never captured a red card or penalty event. However, there are several matches where, after the OOC points, goals and yellow cards are observed.

For the goals, both Goals Home Change and Goals Away Change are examined:

- When there are 2 Goals Home Change events happening after the OOC point, one occurs above the upper control limit and the other below the lower control limit. This means they do not follow a similar pattern, so no conclusion can be drawn from these two points.
- For Goals Away Change, only one data point is present, so no conclusion can be drawn.

For the yellow cards, when looking at both Yellow Cards Home Change and Yellow Cards Away Change, there are a total of 6 OOC points, which provides a promising amount of data to examine. For these points first a hypothesis is drawn and later the data points examined to see if one can accept the hypothesis. Following is the 3-step methodology for yellow card examination:

- It would be expected that if there is a pattern in these OOC points: if there is an upper-type OOC point (indicating an increase in attacks in favor of the home team, since I am looking at the home and away attack differences), followed by a yellow card for the

home team, and if there is a lower-type OOC point (indicating a shift towards increasing attacks from the away team), then a yellow card should follow for the away team.

- When these 6 yellow card events are examined, after 2 upper-type OOC points, the home team receives a yellow card. After 2 lower-type shifts, the away team receives a yellow card. Hence, 4 out of the 6 yellow card events follow a similar pattern. One could hypothesize that a mean shift in the attack metric can provide information about yellow card events. If a team increases its attacks compared to its rival, there is a chance that the team will receive a yellow card.
- However, since 2 of these 6 points behave contrary to the hypothesis, these two OOC points should be examined more closely to see if the behavior can be attributed to another cause. These 2 points correspond to the matches with fixture IDs 19134492 and 19155137. Their control charts are examined in section 5 to find potential alternative causes.

5. Special Cases

For the first instance of the 2 contradicted match, control chart for the match with ID 19134492 is given in Figure 16 to analyze the counter behavior.

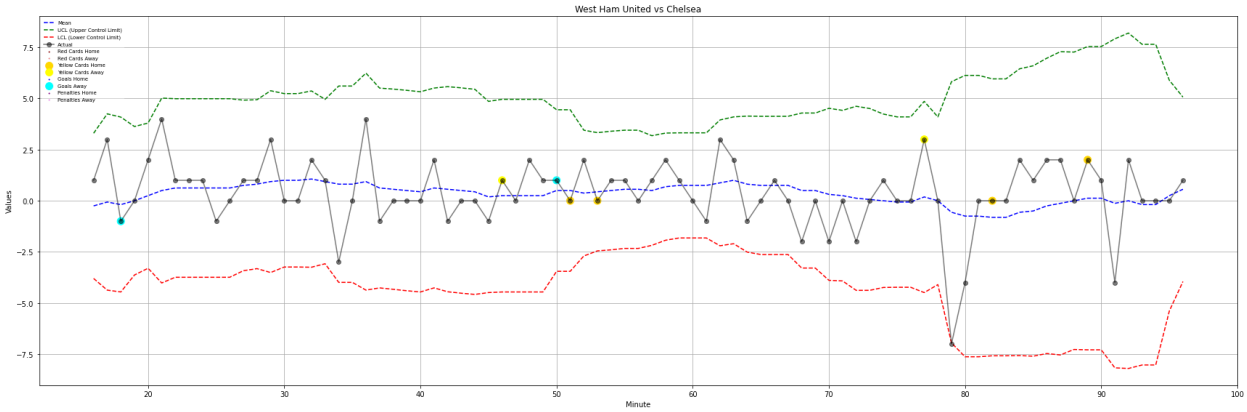


Figure 16

For this match, the OOC point was at the 79th minute, and as can be seen, the mean shifts downward, indicating an increase in attacks from the away team. According to my hypothesis, the away team should receive a yellow card afterward. However, contrary to the hypothesis, the home team receives a yellow card. When examining the minutes around the OOC point, it can be seen that the away team received a yellow card 2 minutes before. This significant event might have caused the behavior of the home team. It is possible that after an aggressive move from one team that results in a yellow card, the other team may become more aggressive and receive a yellow card, even though they did not increase their attack performance.

Since there is an assignable cause/event in this match, it cannot be said that this example invalidates my hypothesis.
Another OOC point that had a counter behavior compared to the hypothesis was for the match with fixture ID 19155137 and its dynamic control chart can be seen in Figure 17 for the analysis of the root cause of the out-of-control point.

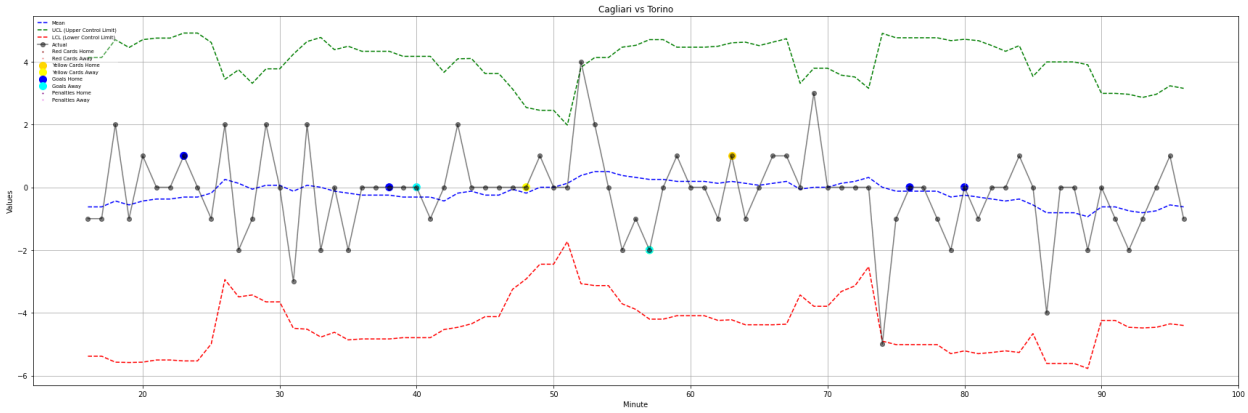


Figure 17

In this match, there are two OOC points, but the one that raises a question about the hypothesis is the one at the 52nd minute. When examining the events around the 52nd minute, there are two significant events:

A goal from away team:

- This goal event happened at the same time as the yellow card I am examining. This suggests that the OOC point may have been an early warning signal for the goal event, not the yellow card that has a misleading pattern for my hypothesis. Hence, this point can also be eliminated. (Since both events occur at the same time in the chart, I cannot see the yellow card in the 57th minute. However, the outlier table given in Figure 15 shows that they happened simultaneously.)

A yellow card for the home team:

- Similar to the previous OOC point that caused behavior contrary to the hypothesis, the home team received a yellow card 3 minutes prior to the OOC point. Therefore, regardless of the OOC point, the home team's aggressive behavior and yellow card may have been the reason for the away team to behave aggressively and receive a yellow card, even though they didn't increase their attacks. Hence, this point is also not significant enough to conclude that my hypothesis is wrong.

- Another thing to mention is that in both OOC points that violate my hypothesis, a yellow card from the opposite team is found, which also supports the idea that this can be a reasonable assignable cause.

6. Similar Pattern Cases

Four control charts that follow a similar pattern and support the hypothesis that an increase in attacks by one team will result in a yellow card for the respective team are plotted below from Figure 18 to 21.

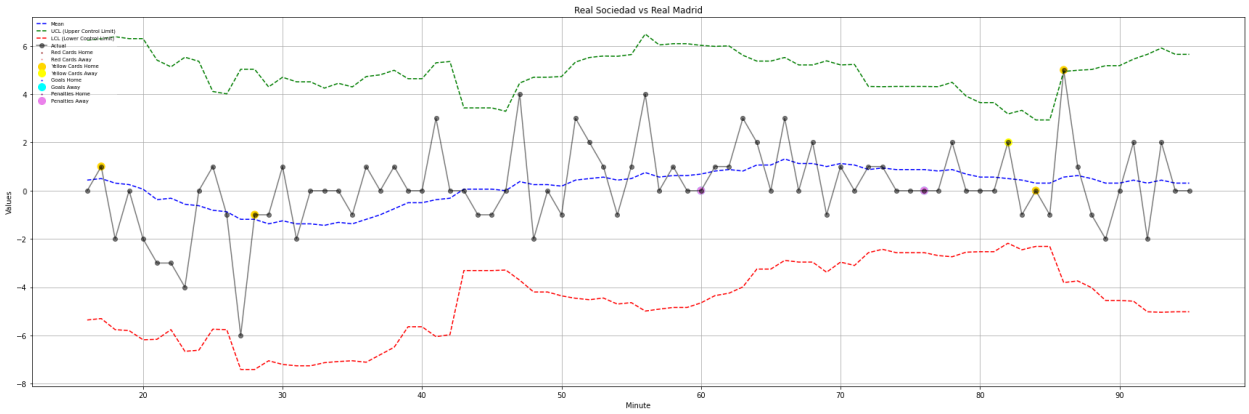


Figure 18

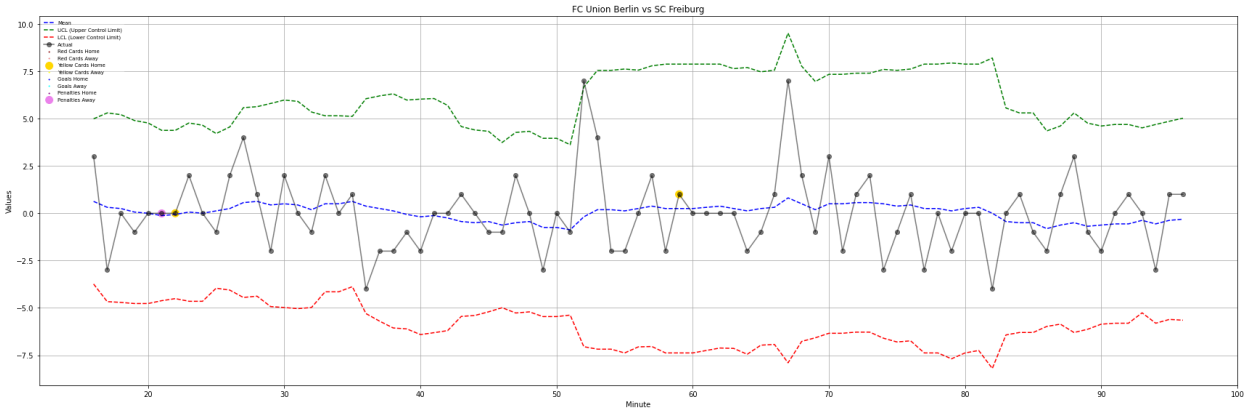


Figure 19

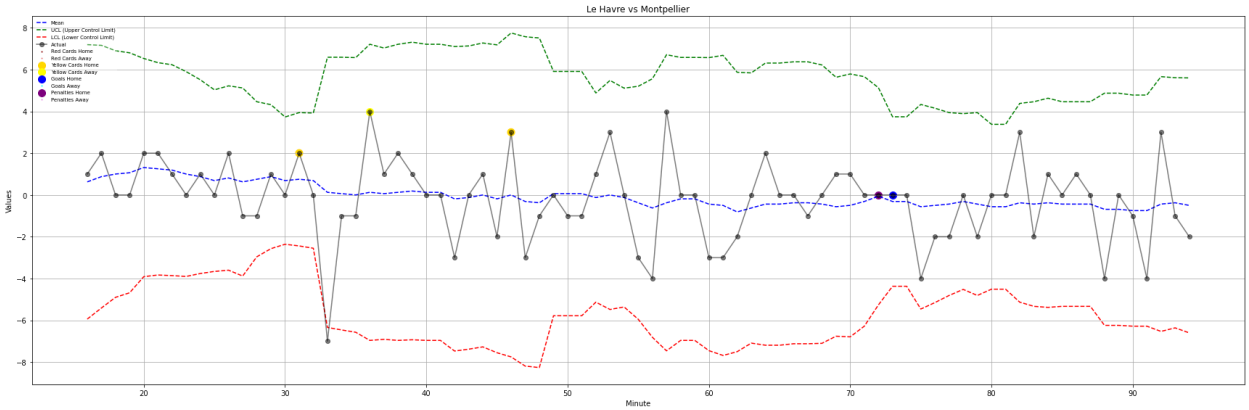


Figure 20

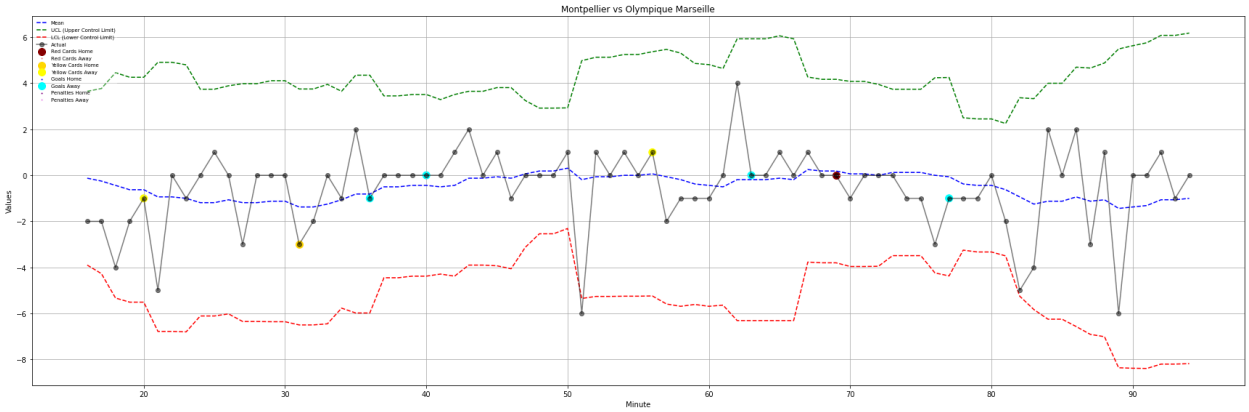


Figure 21

7. Evaluation of Hypothesis

Next step is to define metrics that will evaluate my hypothesis. For this purpose 2 metrics are choosen:

- **Time until detection:** This metric takes the average time difference between the OOC point and the yellow card event that occurred afterwards to evaluate how fast it can detect shift in mean.
- **False alarm rate:** This metric calculates the False Positives meaning that there is an OOC point however there isn't the event that my hypothesis expects.
    - This will be calculated by the following formula: False Positives / Total warnings

Average Time Until Detection:

From the calculation, by using the time difference between the 4 OOC points and the resulting yellow card event, the average time until detection is found to be 3.75 minutes.

False Alarm Rate:

The false alarm rate can be calculated in two ways:

1. Including all OOC points
   - Among the 6 upward mean shifts, only 2 resulted in a yellow card for the home team.
   - Among the 6 downward mean shifts, only 2 resulted in a yellow card for the away team.

   False Alarm Rate = (12-4)/12 = 66.7%

2. Including only the OOC points that resulted in a yellow card:
   - Out of the 6 OOC points that resulted in yellow cards, only 2 were false alarms.

   False Alarm Rate =(6-4)/4 = 33.3%

8. Conclusion

In conclusion, the analysis conducted in this task has demonstrated that the attack difference metric ("Attacks_home - Attacks_away") can be a potentially useful indicator for predicting yellow card events in soccer matches. The association of 6 out of 12 out-of-control (OOC) points with yellow card events across 10 unique matches shows a promising correlation between significant shifts in attack patterns and yellow card actions. The hypothesis that an increase in a team's attacks relative to their opponent may lead to that team receiving a yellow card was supported in 4 out of 6 cases, with the other 2 contradictory cases explainable by other match dynamics. The average time until detection of 3.75 minutes for yellow card events following an OOC point suggests that this method could provide a reasonable early warning system. The false alarm rate that is calculated in section 7 varied depending on the calculation method indicating that there is room for further refinement of the method. The approach did not show significant predictive power for other events such as goals, red cards, or penalties, likely due to the rarity of these events in the dataset. The 15-minute rolling interval approach proved effective in creating dynamic control charts and capturing meaningful shifts in attack patterns throughout the matches. Case studies of contradictory instances revealed that contextual factors, such as prior yellow cards or simultaneous goal events, can influence the relationship between attack patterns and subsequent yellow cards. Although the results are promising, particularly for predicting yellow card events, the relatively high false alarm rate suggests that further refinement of the method and incorporation of additional contextual factors could improve its predictive power. Especially since other metrics in the dataset, such as ball possession, may exhibit important patterns and have correlated data with the attacks metric, they can be utilized together with the attacks metric in multivariate control charts. This approach accounts for the various events

happening in a match that could trigger a yellow card. However, the suggested method is much more complex and harder to understand. Hence, the method used in this study serves as a good starting point for understanding match behavior.

Appendix

```
# Function to anaylze the attack metric that takes window size and the minutes to analyze after the ooc point as a parameter so that analyses can be repeated with different conditions
def Analyze_Attack(data, fixture_id,window=15,max_minute=8):

    fixture_data = data[data["fixture_id"] == fixture_id]
    last_minute=max(fixture_data["minute"])-1
    outlier=[]

    # Calculation of UCL/LCL of each 15 minute rolling interval so that a dynamic control chart is constructed
    for minute in range(window+1,last_minute+1):
        fixture_data_window = fixture_data[(fixture_data["minute"] >= minute-window) & (fixture_data["minute"] <= minute)]

        mean = fixture_data_window["Attacks_difference"].mean()
        std_dev = fixture_data_window["Attacks_difference"].std()

        upper_bound = round(mean + 3 * std_dev,2)
        lower_bound = round(mean - 3 * std_dev,2)

        # When a out of control point found above the UCL, the minute for the OOC point, the changes in events for the following 8 minutes and the time until detecting the events are re
        if fixture_data_window.iloc[-1]["Attacks_difference"]>upper_bound:
            arr=[int(fixture_id),"Upper",minute]
            bools=[0,0,0,0,0,0,0,0]
            minutes=[0,0,0,0,0,0,0,0]
            for imin in range(minute,minute+max_minute):
                if imin>last_minute:
                    break
                if fixture_data.loc[fixture_data["minute"] == imin, "Goals_home_Change"].iloc[0] and bools[0]==0:
                    bools[0]=1
                    minutes[0]=imin-minute
```

A.1. Analyze attack function to examine the matches in 15 min rolling intervals

```
# Function to check the events occured before the out of control situations
def check_total_events_before_ooc_wrt_sign(match_data, out_of_control_data, event_columns, window_size):

    # Initialize dictionaries to store total counts for each sign and for each event
    totals_by_sign = {sign: {event: 0 for event in event_columns} for sign in ['+UCL', '-LCL']}
    total_out_of_control_counts = {'+UCL': 0, '-LCL': 0}

    for fixture_id, group in match_data.groupby('fixture_id'):
        if fixture_id in out_of_control_data:
            out_of_control_df = out_of_control_data[fixture_id]
        else:
            continue

        group = group.sort_values(by='minute')

        # Calculate the events occurred in each minute by taking differencing since the events are cumulatively given
        event_diffs = group[event_columns].diff().fillna(group[event_columns])

        # Set the first row of the differenced series to 0 (ignore the first event at minute 1)
        event_diffs.iloc[0] = 0

        # Replace cumulative data with per-minute counts
        group[event_columns] = event_diffs.clip(lower=0)

        # Count events for each sign (+UCL and -LCL)
        for sign in ['+UCL', '-LCL']:
            # Filter out-of-control situations with respect to sign
            relevant_ooc_minutes = out_of_control_df[out_of_control_df['out_of_control_sign'] == sign]['out_of_control_minutes'].tolist()

            # Increase the total count of out-of-control situations for this sign
            total_out_of_control_counts[sign] += len(relevant_ooc_minutes)
```

A.2. Function for analyzing the events before each OOC points
code